

Chaos Language Algorithm: Standard Attractor Benchmark Summary

ZeroBot / ProtoCosmoBot

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1 Overview

This report summarises the Chaos Language Algorithm (CLA) pure-Python MVP work on five standard chaotic dynamical systems. The CLA pipeline comprises: trajectory generation, fixed-partition M1 symbolization, n-gram chunk mining with non-overlapping rewrite and exact expansion, deterministic context category proposal (exact or JS-divergence clustered), greedy MDL acceptance loop, JSON persistence with edit-log replay, and stable text rendering.

All code is in `projects/chaos-language-algorithm/repos/chaoslang` (GitHub: `bgoertzel-sing/chaos-lang` branch `main`, latest commit `847390c`. All 64 unit tests pass.

2 Systems Benchmark Suite

Five standard chaotic systems were implemented as deterministic, dependency-free Python generators:

2.1 Logistic Map

One-dimensional discrete map:

$$x_{t+1} = r x_t(1 - x_t), \quad r = 4.0, x_0 = 0.123 \quad (1)$$

Fully chaotic at $r = 4$. Produces 4 unique M1 symbols (4 bins on $[0, 1)$). The CLA discovers 8 chunk rules including nested hierarchies ($N_0 \rightarrow N_3 N_9$, $N_0 \rightarrow N_4 N_5$). MDL score: 113.2 bits (0.44 bits/symbol). Exact reconstruction verified.

2.2 Lorenz-63

Three-dimensional continuous-time flow:

$$\dot{x} = \sigma(y - x) \quad (2)$$

$$\dot{y} = x(\rho - z) - y \quad (3)$$

$$\dot{z} = xy - \beta z \quad (4)$$

with $\sigma = 10$, $\rho = 28$, $\beta = 8/3$, $dt = 0.01$, RK4 integration, initial condition $(1, 1, 1)$.

The classic strange attractor. Produces 21 unique M1 symbols from the 3D trajectory. CLA discovers 8 chunk rules with nested references. MDL score: 187.2 bits (0.73 bits/symbol). Exact reconstruction verified.

2.3 Rössler

Three-dimensional continuous-time flow:

$$\dot{x} = -y - z \quad (5)$$

$$\dot{y} = x + ay \quad (6)$$

$$\dot{z} = b + z(x - c) \quad (7)$$

with $a = 0.2$, $b = 0.2$, $c = 5.7$, $dt = 0.05$, RK4 integration, initial condition $(0.1, 0, 0)$.

Simpler attractor structure than Lorenz-63 (single-band fold). Produces 24 unique M1 symbols. CLA discovers 8 chunk rules. MDL score: 189.2 bits (0.74 bits/symbol). Exact reconstruction verified. Similar complexity to Lorenz-63 as expected.

2.4 Mackey-Glass

Scalar delay-differential equation:

$$\frac{dx}{dt} = \beta \frac{x_\tau}{1 + x_\tau^n} - \gamma x \quad (8)$$

with $\beta = 0.2$, $\gamma = 0.1$, $\tau = 17$, $n = 10$, $dt = 0.1$, Euler method with discrete delay buffer, initial history $x = 0.5$.

Delayed-feedback chaotic system; the closest textbook analog to OmegaSim-style delayed-coupling dynamics. Produces only 4 unique M1 symbols (scalar trajectory, narrow dynamic range with these parameters). CLA discovers 6 chunk rules including nested hierarchy ($N_0 \rightarrow N_1 \rightarrow N_5$). Best compression ratio: 0.17 bits/symbol. MDL score: 44.4 bits. Exact reconstruction verified.

2.5 Lorenz-96

N -dimensional continuous-time flow:

$$\dot{x}_i = (x_{i+1} - x_{i-2})x_{i-1} - x_i + F, \quad i = 0, \dots, N-1 \pmod{N} \quad (9)$$

with $F = 8.0$, $dt = 0.01$, RK4 integration. Dimension inferred from initial condition. Default: $N = 5$ with initial $[8.0, 8.0, 8.0, 8.0, 8.0]$.

Chaotic for $F \geq 4$ and $N \geq 4$. This is the primary scalability benchmark because dimension is freely configurable.

3 Dimensionality Scaling Results (Lorenz-96)

Benchmark runs at dimensions $N \in \{5, 8, 12, 16, 20, 24\}$ with 4 bins per axis, 256 post-discard steps, 8 MDL iterations:

Key observations:

- Unique symbols grow rapidly with dimension: 4^N possible symbols, so at $N = 16$ there are $\sim 4.3 \times 10^9$ possible symbols versus 256 actual observations. Most are unique.
- MDL score approaches the raw-entropy ceiling ($\log_2 256 = 8$ bits/symbol is the trivial upper bound; actual unique-symbol entropy approaches ~ 8 bits/symbol at high N).
- Chunk rule count drops at high dimension because few repeated n-grams exist in 256 steps. Longer trajectories or coarser binning would help.

Dim	Unique Symbols	Rules	MDL Score (bits)	Bits/Symbol
5	56	8	234.2	0.91
8	78	8	229.2	0.90
12	107	6	241.4	0.94
16	121	3	248.7	0.97
20	127	6	251.4	0.98
24	148	5	251.5	0.98

Table 1: Lorenz-96 dimensionality scaling. At $\text{dim} \geq 16$, the symbol stream approaches near-incompressibility with 4 bins and 256 steps: most symbols are unique. CLA maintains exact reconstruction at all tested dimensions.

- At $\text{dim} = 20$, the experiment notes state this is comparable to OmegaSim starter-vector dimensionality (~ 17 state variables per hive, $\sim 20\text{--}30$ when including cross-hive coupling fields).
- At $\text{dim} = 24$, 58% of symbols are unique, yet CLA still finds 5 repeatable chunk rules and maintains exact reconstruction.

4 CLA Pipeline Architecture

4.1 Trajectory Generation

Five deterministic generators in `benchmarks/attractors.py`: `logistic_map`, `lorenz63`, `rossler`, `mackey_glass`, `lorenz96`. Continuous systems use fixed-step RK4; Mackey-Glass uses Euler with discrete delay buffer; logistic map is a direct discrete iteration. All are seed-deterministic (no RNG).

4.2 M1 Symbolization

Fixed rectangular partition: for each trajectory point $\mathbf{x} \in \mathbb{R}^d$, each axis i is binned into B equal-width bins over $[\min_i, \max_i]$. The symbol is a concatenated label: `m1:d0_2|d1_0|d2_3` for a 3D point. Bounds are inferred once from the full trajectory (or supplied externally for cross-trajectory comparison). Values outside supplied bounds are clamped to edge bins.

4.3 Chunk Mining

N-gram pattern miner (`mining/ngram.py`): scans for repeated n-grams of length $n \in [n_{\min}, n_{\max}]$ with minimum use count. Non-overlapping rewrite replaces occurrences with a new nonterminal token. Pruning removes productions whose nonterminal is no longer referenced after rewrite (fixes dangling-nonterminal bug).

4.4 Context Categories

Two category induction methods:

1. **Exact** (`ContextCategoryInducer`): groups tokens by exact context signature (left/right neighbour distribution). Category occurrences preserve the observed member for exact reconstruction.

2. **JS-divergence** (`JSCategoryInducer`): clusters tokens whose context histograms have Jensen-Shannon divergence below a threshold. Implemented in `categorization/js.py` with greedy sorted-key clustering against centroid. Integrated into category proposal generation as an alternative to exact grouping.

4.5 MDL Scoring

Two-part MDL (`scoring/mdl.py`): model cost = sum of rule overhead (0.8 bits/production) + production RHS length + category overhead (14.0 bits/category) + member set size + edit-log cost (0.1 bits/edit). Data cost = token count for base tokens + $1 - \log_2 p(\text{member}|\text{category})$ for category occurrences. Greedy acceptance: a chunk/category proposal is accepted iff total MDL decreases.

4.6 Persistence

JSON state serialization (`persistence.py`): round-trips `GrammarState` to/from JSON with sorted keys for determinism. Edit-log replay reconstructs state from initial symbols + edit sequence. `grammar_to_text` provides stable human-readable grammar rendering. All round-trips preserve exact reconstruction.

5 Common Benchmark Parameters

All experiments use:

- `steps` = 256 (post-discard)
- `discard` = 64 (transient removal)
- `bins` = 4 (M1 partition granularity per axis)
- `iterations` = 8 (MDL loop iterations)
- `seed` = 0 (for CLA’s internal RNG used in tie-breaking)

6 OmegaSim Dimensionality Context

The current OmegaSim A6 single-hive model has 17 state variables per hive: 4 motivation, 4 fatigue, 4 threshold, plus scalars (artifact, provenance_debt, risk, prediction_error, last_action). A three-hive A8 configuration would have ~ 51 state variables plus inter-hive coupling fields.

The CLA Lorenz-96 scaling experiments at $\text{dim} = 20\text{--}24$ are directly comparable to single-hive OmegaSim dimensionality. At these dimensions CLA maintains exact reconstruction but approaches raw-entropy compression limits with 4 bins and 256 steps.

The next step is selecting toy attractor systems with dimensionality $\sim 17\text{--}30$ to bridge from textbook benchmarks to OmegaSim-scale traces, and testing whether CLA can recover known grammatical structure at that scale with longer trajectories and/or coarser binning.

System	Dim	Unique Symbols	Rules	Bits/Symbol
Logistic Map	1	4	8	0.44
Mackey-Glass	1	4	6	0.17
Lorenz-63	3	21	8	0.73
Rössler	3	24	8	0.74
Lorenz-96 (5)	5	56	8	0.91
Lorenz-96 (8)	8	78	8	0.90
Lorenz-96 (12)	12	107	6	0.94
Lorenz-96 (16)	16	121	3	0.97
Lorenz-96 (20)	20	127	6	0.98
Lorenz-96 (24)	24	148	5	0.98

Table 2: All 5 standard attractor systems with Lorenz-96 dimensionality scaling. All runs: 256 steps, 4 bins, 8 iterations, exact reconstruction verified.

7 Summary Table

8 Repository State

- Repository: `bgoertzel-sing/chaos-language-algorithm`
- Branch: `main`
- Latest commit: `847390c` (persistence complete)
- Tests: 64 passing (Python 3.9 on MacBook)
- Dependencies: pure Python stdlib (no numpy, scipy, or torch)
- Sprint tasks 1–5 complete: symbolic MVP, M1 attractor controls, benchmark experiments, JS-divergence clustering, persistence.